

Submission Summary

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International Conference on Information Management 2025

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Paper Title

LINE Bot結合大型語言模型之英語學習系統設計與實作(Design and Implementation of an English Learning System Integrating LINE Bot with Large Language Models)

Abstract

本研究設計並實作了一個基於LINE Bot的英語學習系統，整合大型語言模型（LLM）、語音轉文字（Whisper）及文字轉語音（gTTS）技術，提供互動式英語學習環境。系統以Flask框架為基礎，實現了英語對話練習、單字造句及跟讀練習三大核心功能。研究特別關注提示工程（Prompt Engineering）在教育情境的應用，設計了符合互動模式與回饋機制。使用者回饋顯示，此系統能提供即時、個性化的語言學習體驗，有效降低學習英語的門檻，提升英語互動能力。本研究為人工智慧在語言教育中的應用提供了實證案例，對促進英語學習具有參考價值。

關鍵詞：英語學習、LINE Bot、大型語言模型、提示工程

This research designs and implements an English learning system based on LINE Bot, integrating Large Language Models (LLM), speech-to-text (Whisper), and text-to-speech (gTTS) technologies to provide an interactive English learning environment. The system is built on the Flask framework and implements three core functions: English conversation practice, word sentence construction, and read-aloud practice. The research particularly focuses on the application of Prompt Engineering in educational contexts, designing prompts that align with interactive patterns and feedback mechanisms. User feedback indicates that the system provides immediate, personalized language learning experiences, effectively lowers the barrier to learning English, and enhances English interaction abilities. This research provides an empirical case for the application of artificial intelligence in language education and offers reference value for promoting English learning.

Keywords: English Learning, LINE Bot, Large Language Models, Prompt Engineering

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Submission Files

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